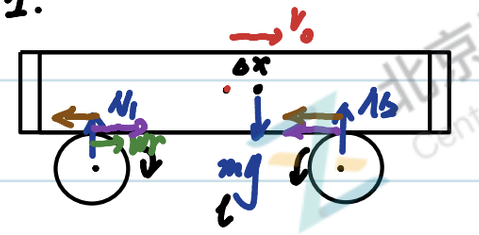
新 → 老 → 变.

0 → 会做 → 做对. → 拿全.

不同策略.
不同难度.
不同题型

29套.

1.



(1). $N_1+N_2=mg$.

$N_1(\frac{l}{2}+\Delta x)=N_2(\frac{l}{2}-\Delta x)$.

$\Rightarrow N_1=\frac{\frac{l}{2}-\Delta x}{l}mg$

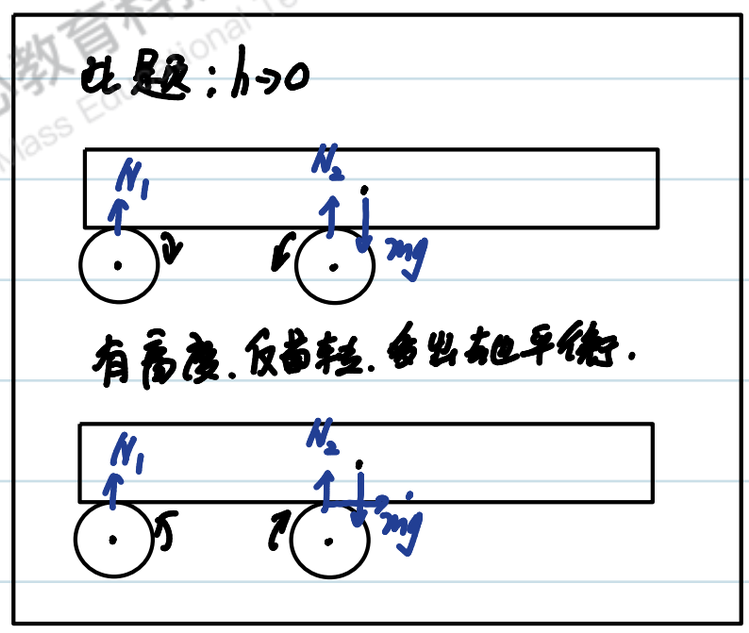
$N_2=\frac{\frac{l}{2}+\Delta x}{l}mg$.

$F=\mu N$

$\Rightarrow \Delta F=-\mu N_2+\mu N_1=-\frac{2\mu mg}{l}\Delta x$.

$\Rightarrow T=2\pi\sqrt{\frac{l}{2\mu g}}$.

(2). l 较小. v_0 较小. 简谐 (已知 l. 求 v_0).



$\frac{1}{2}mv_0^2=\frac{1}{2}k(\frac{l}{2})^2$

$\Rightarrow v_0=\sqrt{\frac{\mu gl}{2}}$.

条件. $v_0<\omega r$.

$\Rightarrow l<\frac{2\omega^2r^2}{\mu g}$.

l 较大. 先匀减. 再简谐.

条件. $l>\frac{2\omega^2r^2}{\mu g}$.

$a=\mu g$.

设. 偏离 x 时. 进入简谐. 此时速度 ωr

匀减: $2ax=v_0^2-\omega^2r^2$.

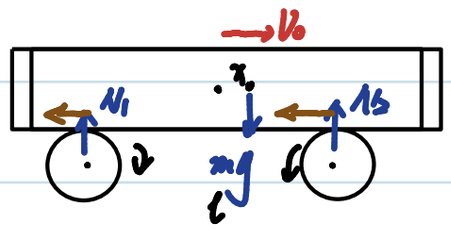
简谐. 能: $\frac{1}{2}m(\omega r)^2+\frac{1}{2}kx^2=\frac{1}{2}k(\frac{l}{2})^2$

$\Rightarrow \frac{2\mu}{g}\omega^2r^2+x^2=\frac{l^2}{4}$

$\Rightarrow \frac{v_0^2-\omega^2r^2}{2\mu g}=\sqrt{\frac{l^2}{4}-\frac{1}{2\mu g}\omega^2r^2}$.

$\Rightarrow v_0=\sqrt{(\mu^2g^2l^2-2\mu g(\omega^2r^2)^{\frac{1}{2}}+\omega^2r^2}$.

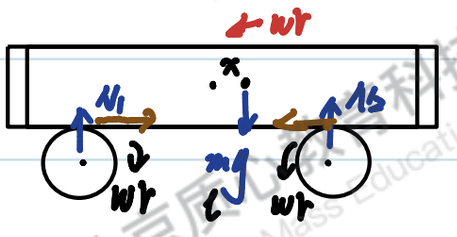
(3). 匀减.



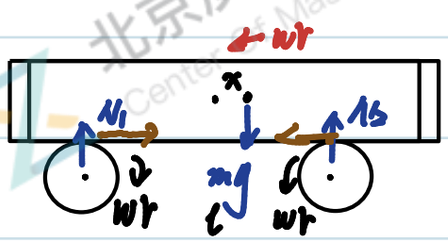
一种理解
(已知: v_0.. $A=\frac{l}{2}$).

另一种理解
(已知: $A=\frac{l}{2}$. 不知 v_0).

简谐



匀速.

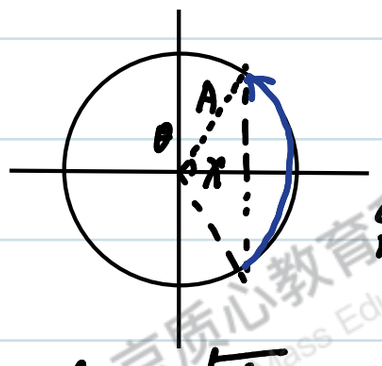


左. 滑动. $f_1=\mu N_1=\mu g$

右. 静摩擦 $f_2\le \mu N_2 g$.

匀减. $t_1=\frac{v_0-\omega r}{\mu g}$.

简谐.



相: $\cos\theta=\frac{x}{A}$.

$t_2=\frac{2\theta}{2\pi}2\pi\sqrt{\frac{l}{2\mu g}}$.

能: $\frac{1}{2}m(\omega r)^2+\frac{1}{2}kx^2=\frac{1}{2}kA^2$.

$\Rightarrow t_2=2\sqrt{\frac{l}{2\mu g}}\arccos\frac{v_0^2-\omega^2r^2}{\sqrt{2\mu g(l\omega^2r^2+(v_0^2-\omega^2r^2)^2}}$

匀速.

$t_3=\frac{x}{\omega r}$.

$t_{总}=t_1+t_2+t_3=\frac{v_0-\omega r}{\mu g}+2\sqrt{\frac{l}{2\mu g}}\arccos\frac{v_0^2-\omega^2r^2}{\sqrt{2\mu g(l\omega^2r^2+(v_0^2-\omega^2r^2)^2}}+\frac{v_0^2-\omega^2r^2}{2\mu g\omega r}$.

2. m_1 ---- r ---- m_2 ---- $2r$ ---- m_3

m_2: $m_2a_2=\frac{Gm_2m_3}{(2r)^2}-\frac{Gm_2m_1}{r^2}$

$\Rightarrow a_2=\frac{Gm_3}{4r^2}-\frac{Gm_1}{r^2}$.

以 m_2 为参.

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m_3:

$$m_3a_3 = \frac{Gm_1m_3}{(3r)^2} + \frac{Gm_2m_3}{(2r)^2} + m_3a_2$$

m_1:

$$m_1a_1 = \frac{Gm_2m_1}{(3r)^2} + \frac{Gm_3m_1}{r^2} - m_1a_2.$$

约束: $a_3 = 2a_1$

$$\Rightarrow \frac{2}{9}m_3 + 2m_2 - \frac{2}{9}m_3 + 2m_1$$

$$= \frac{1}{9}m_1 + \frac{1}{4}m_3 + \frac{1}{4}m_2 - m_1$$

$$\Rightarrow m_3 = \frac{108m_1 + 63m_2}{19}.$$

(2). $a_m = -\frac{GM}{r^2}$

$t = \frac{1}{2}T$. 类比.

$T = 2\pi\sqrt{\frac{R^3}{GM}}$.



m_1: $m_1a_1 = \frac{Gm_3m_1}{(3r)^2} + \frac{Gm_2m_1}{r^2} - a_2m_1$

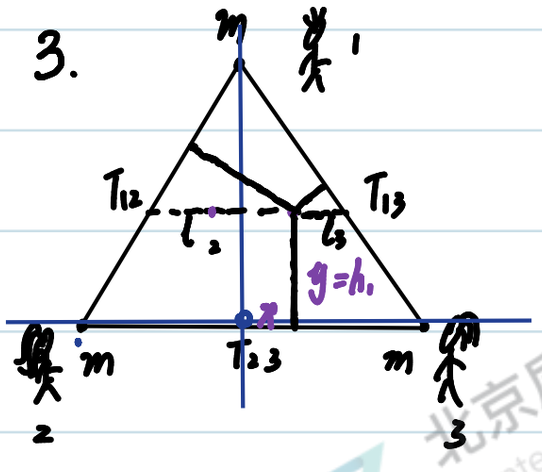
$$\Rightarrow a_1 = -G\left(\frac{41}{171}m_1 + \frac{41}{76}m_2\right)\frac{1}{r^2}$$

$GM \to G\cdot\left(\frac{41}{171}m_1 + \frac{41}{76}m_2\right)$.

$a \to \frac{r}{2}$

$$\Rightarrow t = \pi\sqrt{\frac{r^3}{8G\left(\frac{41}{171}m_1 + \frac{41}{76}m_2\right)}}$$

3.



对每个人. 临界. 当两个人合力拉不动.

不妨分析 1. 的临界.

$$(a_1mg)^2 = (a_2mg)^2 + (a_3mg)^2 + 2a_2mg\cdot a_3mg\cdot\cos 60°$$

$$\Rightarrow a_1^2 = a_2^2 + a_3^2 + a_2a_3$$

直角坐标. 详细算一下.

如图: 设 $h_i = \frac{a_i}{a_1+a_2+a_3}$

几何: $h_2 - h_3 = \frac{\sqrt{3}}{3}(l_2 - l_3)$, $l_2 - l_3 = 2x$

上式: $h_1^2 = (h_2+h_3)^2 - h_2h_3$.

数学: $4h_2h_3 = (h_2+h_3)^2 - (h_2-h_3)^2$

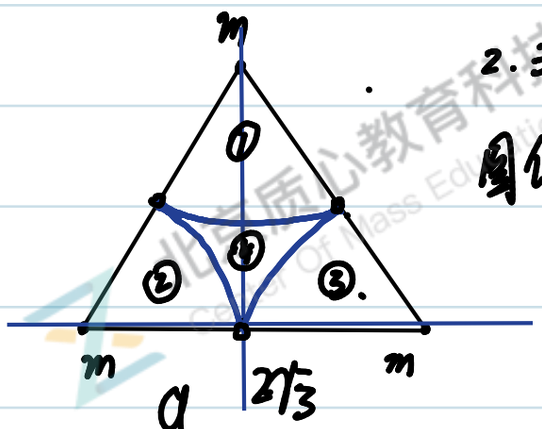
$$= (1-y)^2 - 3\frac{x^2}{3}.$$

$$\Rightarrow y^2 = (1-y)^2 - \frac{1}{4}\left[(1-y)^2 - \frac{3x^2}{3}\right].$$

$$\Rightarrow \left(\frac{y+\frac{1}{3}}{2/3}\right)^2 - \left(\frac{x}{2}\right)^2 = 1.$$

双曲线. $a = 2/3$. $b = 2$. $c = \sqrt{a^2+b^2} = 4$.

中心. $(0,-3)$. 焦点. $(0,4)$. 的双曲线的一部分.



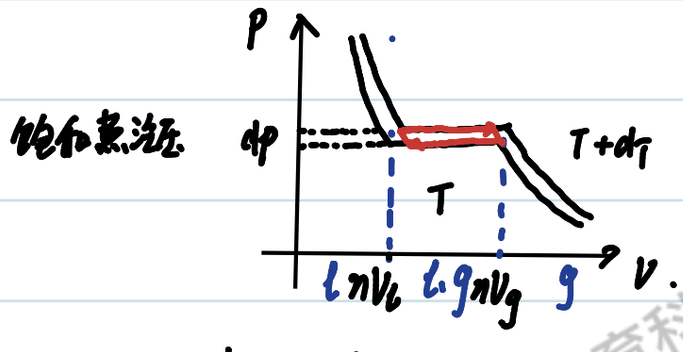
2. 3. 3 滑. 3 滑. 轮换对称.

图像. 旋转 120°.

①. 1 安全. ②. 2 安全 ③ 3 安全. ④. 都不安全.

4. 真. 克拉伯龙方程.

$$\frac{dp}{dT} = \frac{\lambda_m}{V_g - V_l}\frac{1}{T}$$



构建一个. 小卡诺.

$$\gamma = \frac{dT}{T} = \frac{dW}{Q}.$$

$dW = dp\cdot(nV_g - nV_l)$

$Q = n\lambda_{mol}$.

$\because V_l \ll V_g$, λ_{mol}. 为定值.

理解: ① V_g 是常数

② V_g 是变量. 重新、

积分一遍.

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$\Rightarrow \frac{dT}{T} = \frac{dp(nV_g - nV_L)}{n \cdot \lambda_{mol}}$

$\Rightarrow \frac{dp}{dT} = \frac{\lambda_{mol}}{V_g - V_L} \cdot \frac{1}{T}$

当 $V_L << V_g$ 时

$\Rightarrow \frac{dp}{dT} = \frac{\lambda_{mol}}{TV_g}$

$\Rightarrow p = p_0 + \frac{\lambda_{mol}}{V_g} \ln \frac{T}{T_0}$

(2). $V_L < V_S$.

$\frac{dp}{dT} < 0$.

持 $\Rightarrow p \nearrow \Rightarrow T \searrow \Rightarrow$ 骨架液化.

过 $\Rightarrow p \searrow \Rightarrow T \nearrow \Rightarrow$ 骨架.相变成固态.

(3). $m = 9kg$.

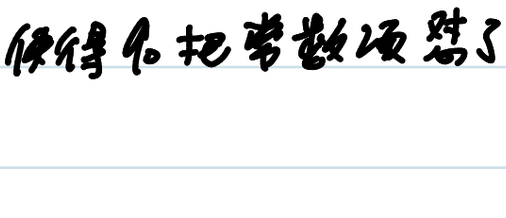
$\Rightarrow V_L = 9L$. $V_2 = 10L$.

$Q = 2.16 \times 10^5 J$. $T = 310K$.

代入 $\frac{dp}{dT} = \frac{dp(V_L - V_2)}{Q}$.

$\Rightarrow dp = 6.97 \times 10^5 Pa$.

5.



(1) $W_\varepsilon = Q_1 + Q_2 = 8.43mJ$.

$W_\varepsilon = \varepsilon \cdot Q_\varepsilon$

$\Rightarrow Q_\varepsilon = \frac{W_\varepsilon}{\varepsilon} = 2.108mC$.

$Q_2 = \frac{1}{2}\frac{Q_C^2}{C}$

$\Rightarrow Q_C = \sqrt{2CQ_2} = 2mC$.

$\Rightarrow Q_R = Q_\varepsilon - Q_C = 0.108mC$.

(2). 充电：是一个线性微分方程.

有常数项. 设 $q(t) = q'(t) + q_0$.

使得 q_0 把常数项消了.

$-\frac{dq'}{dt} \cdot \frac{R \cdot r}{R+r} = \frac{q'}{C}$. $\Rightarrow q' = \square e^{-\frac{t}{\tau}}$

$\tau = \frac{R \cdot r}{R+r} \cdot C = 75s$.

$\Rightarrow q = \square e^{-\frac{t}{\tau}} + \Delta$.

放电电路.

$-\frac{dq}{dt} R = \frac{q}{C}$. $\Rightarrow q = \square e^{-\frac{t}{\tau'}}$

$\tau' = R \cdot C = 300s$.

$\Delta t_1, \Delta t_2 << \tau, \tau'$. 可以认为充放电时电容上电荷不变. 设为 Q

则充电电荷=放电电荷.

充: $\Delta Q_1 = (\frac{\varepsilon - \frac{Q}{C}}{r} - \frac{Q}{C}) \cdot \Delta t_1$

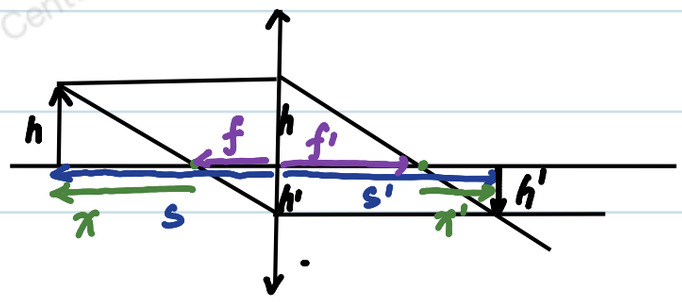
放: $\Delta Q_2 = \frac{Q}{R} \cdot \Delta t_2$.

则: $\Delta Q_1 = \Delta Q_2 = \Delta Q$

$\Rightarrow Q = \frac{\varepsilon \cdot C \cdot R}{3r + R} = 4mC$. $\Rightarrow \Delta Q = 0.004mC$.

$W_\varepsilon = \varepsilon \cdot \Delta Q = 0.016mJ$

6.



$\frac{h'}{h} = \frac{f}{x}$ $\frac{h'}{h} = \frac{x'}{f'}$

$\Rightarrow$. 牛顿: $xx' = f \cdot f'$.

$\frac{h'}{h} = -\frac{f}{x} = -\frac{x'}{f'}$

比题:

(1). $\gamma_2 = \frac{f^2}{\gamma_1}$. $y_2 = -y_1 \frac{f}{x_1}$

(2). 点→点. 线→线. 面→面.

二项式→二项式.

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$(x_1-h)^2+y_1^2=r^2$.

$\Rightarrow \left(\frac{f^2}{x_2}-h\right)^2+\left(-y_2\cdot\frac{f}{x_2}\right)^2=r^2$.

$\Rightarrow (h^2-r^2)\cdot x_2^2-2hf^2\cdot x_2+f^2\cdot y_2^2+f^4=0$

当 $h^2-r^2=f^2$ 时为圆.

即 $(x_2-b)^2+y^2=r^2$

(3). $r=\sqrt{h^2-f^2}$.

$x=h+\sqrt{h^2-f^2}\cos\theta$

$\dot{x}_b=-w\sqrt{h^2-f^2}\sin\theta$.

$v_x=\dot{x}_b=-\frac{f^2}{x_1^2}\dot{x}_1$

$=\frac{wf^2\sqrt{h^2-f^2}}{(h+\sqrt{h^2-f^2}\cos\theta)^2}\cdot\sin\theta$.

$y_1=\sqrt{h^2-f^2}\sin\theta$.

$\dot{y}_1=w\sqrt{h^2-f^2}\cos\theta$.

$v_y=\dot{y}_2=\frac{y_1f}{x_1^2}\dot{x}_1-\frac{f}{x_1}\dot{y}_1$

$=\frac{-wy_1f\sqrt{h^2-f^2}\cdot\sin\theta}{(h+\sqrt{h^2-f^2}\cos\theta)^2}-\frac{f\cdot w\sqrt{h^2-f^2}}{h+\sqrt{h^2-f^2}\cos\theta}\cdot\cos\theta$.

(4). 最亮. 横向. 纵向. 放大率最小.

$y_1=0$. $\rightarrow y_2=0$.

$x_1=h+r$. $\Rightarrow x_2=h-r$.

(5). $x_1x_2=f_1f_2$. $\frac{y_2}{y_1}=-\frac{x_2}{f_2}=-\frac{f_1}{x_1}$

$(x_1-h)^2+y_1^2=r^2$.

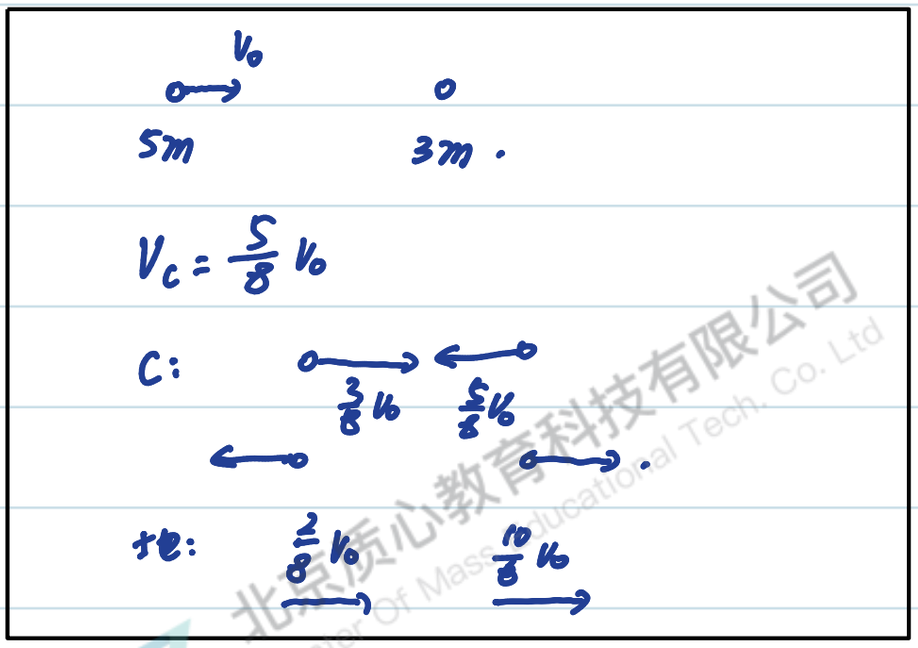
$\Rightarrow (h^2-r^2)x_2^2-2hf_1f_2\cdot x_2+f_2^2\cdot y_2^2+f_1^2f_2^2=0$.

$\Rightarrow h^2-r^2=f_2^2$ 时. 圆.

$\Rightarrow \left(x_2-h\frac{f_2}{f_1}\right)^2+y_2^2=\left(r\frac{f_2}{f_1}\right)^2$

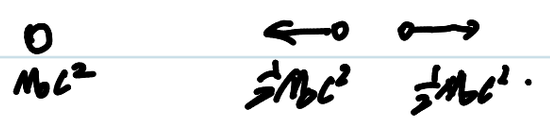
$\Rightarrow h'=h\frac{f_2}{f_1}$ $r'=r\cdot\frac{f_2}{f_1}$.

7.



质心系. 速度 $v_c=v_0$ $\beta=\frac{v_0}{c}$. $\gamma=\frac{1}{\sqrt{1-\beta^2}}$

质心系:



能: $E'=\frac{1}{3}m_0c^2$.

动: $E'^2=(m_0c^2)^2+(p'c)^2$

$\Rightarrow p'=\sqrt{\frac{1}{9}m_0^2c^2-m_0^2c^2}$.

换回地面: $\begin{bmatrix}\frac{E_2}{c}\\ p_2\end{bmatrix}=\begin{bmatrix}\gamma & \beta\gamma\\ \beta\gamma & \gamma\end{bmatrix}\cdot\begin{bmatrix}\frac{E'}{c}\\ p'\end{bmatrix}$

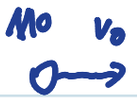
$\Rightarrow \frac{E_2}{c}=\gamma\left(\frac{E'}{c}+\beta\cdot p'\right)$.

$p_2=\gamma(p'+\beta\frac{E'}{c})$.

$\Rightarrow \frac{v_2}{c}=\frac{p_2}{\frac{E_2}{c}}=\frac{p'+\beta\frac{E'}{c}}{\frac{E'}{c}+\beta\cdot p'}$.

$=\frac{\sqrt{\frac{1}{9}m_0^2c^2-m_0^2c^2}+\frac{v_0}{c}\cdot\frac{1}{3}m_0c}{\frac{1}{3}m_0c+\frac{v_0}{c}\sqrt{\frac{1}{9}m_0^2c^2-m_0^2c^2}}$

思考:

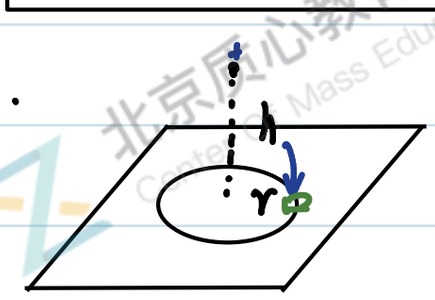


衰变成3个静止能为0的粒子.

动量. 1沿 θ 方向射出.

求: 2方向与水平夹角最大值.

8.



(1). $E=\frac{1}{4\pi\varepsilon_0}\cdot\frac{2qh}{(h^2+r^2)^{3/2}}$

$-E\cdot ds=\sigma\cdot ds/\varepsilon_0$

$\Rightarrow \sigma=-\frac{1}{2\pi}\frac{qh}{(h^2+r^2)^{3/2}}$.

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$$\frac{q}{\varepsilon_0}=\oiint\vec{E}\,d\vec{S}=\oiint\frac{\vec{j}\,ds}{\sigma}=\frac{1}{\sigma}\cdot$$

$$\frac{dq}{dt}\cdot\frac{1}{\sigma}=-\frac{1}{\varepsilon_0}q\cdot$$

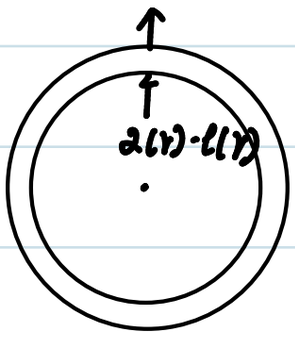
$$\Rightarrow -\frac{dq}{dt}=-\frac{1}{\varepsilon_0\sigma}q$$

$$\Rightarrow q=q_0e^{-\frac{1}{\varepsilon_0\sigma}t}\cdot$$

$$\frac{dq}{dt}=\frac{d\sigma}{dt}\cdot 2\pi r\cdot dr$$

$$=-q\nu r\cdot dr\cdot\frac{r^2-2h^2}{(h^2+r^2)^{5/2}}\cdot$$

$2(r+dr)\cdot l(r+dr)$



$$2(r+dr)\,l(r+dr)-2(r)\,l(r)=\frac{dq}{dt}\cdot$$

$$\Rightarrow\cdot d(2\cdot 2\pi r)=\frac{q\nu}{2}\cdot\frac{r^2-2h^2}{(h^2+r^2)^{5/2}}\cdot d(r^2)$$

$$\Rightarrow 2\cdot 2\pi r=q\nu\cdot\frac{r^2\cdot}{(h^2+r^2)^{3/2}\cdot}$$

$$\Rightarrow\cdot 2=\frac{q\nu}{2\pi}\frac{r\cdot}{(h^2+r^2)^{3/2}}$$

(3). $M=\int\cdot dM=\int BI\,dr\cdot r$

$$=\int\frac{A}{\pi r^2}\cdot 2\cdot 2\pi r\cdot r\cdot dr\cdot$$

$$=\int\cdot 2\pi\frac{Aq\nu}{2\pi}\frac{r\cdot}{(h^2+r^2)^{3/2}}dr\cdot$$

$$=\frac{q\nu A}{2}\int_0^\infty\frac{d(r^2)\cdot}{(h^2+r^2)^{3/2}}\cdot$$

$$=\frac{q\nu A}{h}\cdot$$

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